

CN Lab: Week 4, Socket Programming

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1) TCP Client-Server Communication

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\dhriti> cd .\Downloads\
PS C:\Users\dhriti> cd .\Downloads\
PS C:\Users\dhriti\Downloads> python .\tcpServer.py
Socket successfully created
Socket binded to 12345
Socket is listening...
❖ Got connection from ('127.0.0.1', 54583)
PS C:\Users\dhriti\Downloads>
PS C:\Users\dhriti> cd .\Downloads\
PS C:\Users\dhriti\Downloads> python .\tcpClient.py
Socket successfully created
the socket has successfully connected to google
❖ PS C:\Users\dhriti\Downloads>
```

2) Echo Client-Server Communication

```
PS C:\Users\dhriti\Downloads> python .\echoServer.py
Socket successfully created
Server binded to port 32007
Server is listening...
Connected by ('127.0.0.1', 52444)
At server: Welcome to campus after a long gap!
❖ PS C:\Users\dhriti\Downloads>
PS C:\Users\dhriti\Downloads> python .\echoClient.py
Connected to server
At client: Echo: Welcome to campus after a long gap!
❖ PS C:\Users\dhriti\Downloads>
```

3)UDP Client-Server communication

```
PS C:\Users\dhriti\Downloads> python .\UDPServer.py
UDP server up and listening
Message from Client:b'Hello UDP Server'
❖ PS C:\Users\dhriti\Downloads>
PS C:\Users\dhriti\Downloads> python .\UDPClient.py
Message from Server b'Hello UDP client'
❖ PS C:\Users\dhriti\Downloads>
```

4)UDP Client-Server communication (Upper Case)

```
PS C:\Users\dhriti> cd .\Downloads\
PS C:\Users\dhriti> cd .\Downloads\
PS C:\Users\dhriti\Downloads> python .\UDPUpperCaseServer.py
The server is ready to receive
❖ PS C:\Users\dhriti\Downloads>
PS C:\Users\dhriti> cd .\Downloads\
PS C:\Users\dhriti\Downloads> python .\UDPUpperCaseClient.py
Input lowercase sentence:welcome to cn lab
WELCOME TO CN LAB
❖ PS C:\Users\dhriti\Downloads>
```